# Iftekhirul Islam Muhid

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## Education

**Gopalganj Science and Technology University** March 2022 – October 2026
*Bachelor of Science (B.Sc.) in Electrical and Electronic Engineering* *CGPA: 3.25 out of 4.0*

## Undergraduate Thesis

**A Comprehensive Study of Electrical, Optical, and Optoelectronic Properties of $MgSiP_2$-Based Solar Cell Using Quantum ESPRESSO, SCAPS-1D Simulation and Machine Learning**

- Performed Density Functional Theory (DFT) calculations using Quantum ESPRESSO to determine the electronic and optical properties of Mg-doped $SiP_2$ ($MgSiP_2$).
- Designed a solar cell with $MgSiP_2$ as the absorber layer in SCAPS-1D and optimized the device parameters to improve performance.
- Implemented machine learning models to predict solar cell performance from device parameters, and applied SHAP analysis to identify the features that most strongly influence efficiency.

## Publications

- M. Nayem, N. H. Ratul, **I. I. Muhid**, M. Mohiuzzaman, M. S. Farid, S. A. Dipa, "A Reliability Oriented Ensemble Learning Framework for Cardiovascular Disease Prediction with SHAP Based Explainability," IEEE SPICSCON, 2026. **(Accepted)**

## Research Interests

Machine Learning & Deep Learning · Robotics & UAVs · Embedded Systems

## Skills

**Programming Languages:** C, C++, Python, MATLAB, 8086 Assembly Language; OOP, Data Structures & Algorithms
**Machine Learning & Data Science:** NumPy, Pandas, Matplotlib, Seaborn, Plotly, Scikit-learn, SciPy, TensorFlow, PyTorch
**Automation & Industrial Tools:** TIA Portal, Distributed Control Systems (DCS), MATLAB Simulink
**Computational Physics & Device Simulation:** Quantum ESPRESSO (DFT), SCAPS-1D (solar cell simulation)
**Embedded & Hardware Systems:** Arduino programming
**Simulation & Design:** Proteus, AutoCAD, Logisim
**Developer Tools:** Git, GitHub, Linux & Windows shell scripting, Overleaf LaTeX

## Projects

**Autonomous Line Following Robot** | *Arduino Programming*

- Detects the line in real time with an IR sensor array and chooses the correct track dynamically.
- Uses PID control for smooth sharp turns and minimal deviation.
- Built with Arduino Nano, QTR sensor array, and motor driver.

## Training

**Training Institute for Chemical Industries (TICI), BCIC** Aug. 2025
*Industrial Technology on Electrical Engineering & Instrumentation (2 weeks)* *Grade: A+*

- Covered power systems (transformers, generators, substations, earthing, synchronization) and process instrumentation & automation (control loops, PLC, DCS, SCADA).

## Awards, Teaching & Extra-Curricular

- Ranked 17th position at IEEEXtreme 17.0 in the Bangladesh region
- Instructor of C and Python Programming at the University Programming Club
- President, Programming and Soft Skills Club (2025 – 2026)
- Webmaster, IEEE GSTU Student Branch (2024 – 2025)
- Organized and mentored participants of RoboFest'24 (Intra-University Robotics Competition)
- Participated in national robotics competitions such as Ignition and BITROBOT
- Founder & Director, The Wizzy Corp. (A start-up company, founded in 2024)